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REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.

TESTPAD SYMBOL = TESTPAD

TABLE A

REFERENCE DESIGNATION	POWER INPUT TO PINS							
	+5V							
	+3.3V							
	GND							

DESCRIPTION:

1. THIS PCB USES AN ESP32 TO RECEIVE AN IMAGE THROUGH A NORMAL OR THERMAL CAMERA.
2. THE ESP32 WILL PRE-PROCESS THE IMAGE.
   A. CONVERT TEMPERATURE MAP FROM THERMAL SENSOR TO GRAYSCALE.
   B. CONVERT 1D ARRAY FROM STEP A TO 12 ROWS 16 COLUMNS.
   C. SEND FRAME TO FPGA UTILIZING SPI.
3. FPGA STORES INCOMING PIXELS IN BRAM AS 16X12 IMAGE.
4. THE FPGA APPLIES HAAR FILTERS (LOW PASS & HIGH PASS].
5. FPGA SENDS HAAR FILTER OUTPUT BACK TO ESP32 THROUGH SPI.
6. ESP32 POST PROCESSES THE HAAR OUTPUT AND DISPLAYS TO OLED(S) DISPLAY.

INSTALLATION:

1. MANUALLY LOAD FPGA.BIN TO THE SD CARD.
   A AT THE START OF ESP32 CODE, IT WILL SEND A BITSTREAM FROM THE SD CARD TO THE FPGA.
   THIS WILL GIVE THE FPGA THE INSTRUCTIONS TO DO HAAR FILTERS.
2. INSTALL ALL HEADER PINS WITH SOLDERING IRON.
3. CONNECT PWR SW+FUSE, OLEDS, AND MAX90641 TO THEIR ASSIGNED HEADER PINS.
4. UPLOAD SOFTWARE TO ESP32 BOARD.

CONTRACT NO	
DWN BY	
CHECK BY	F ABUEIDA
APPV BY	PROF F CHUNG
MKS Rev MKS Date	

CALIFORNIA STATE UNIVERSITY - LONG BEACH
ELECTRICAL ENGINEERING
LONG BEACH, CALIFORNIA

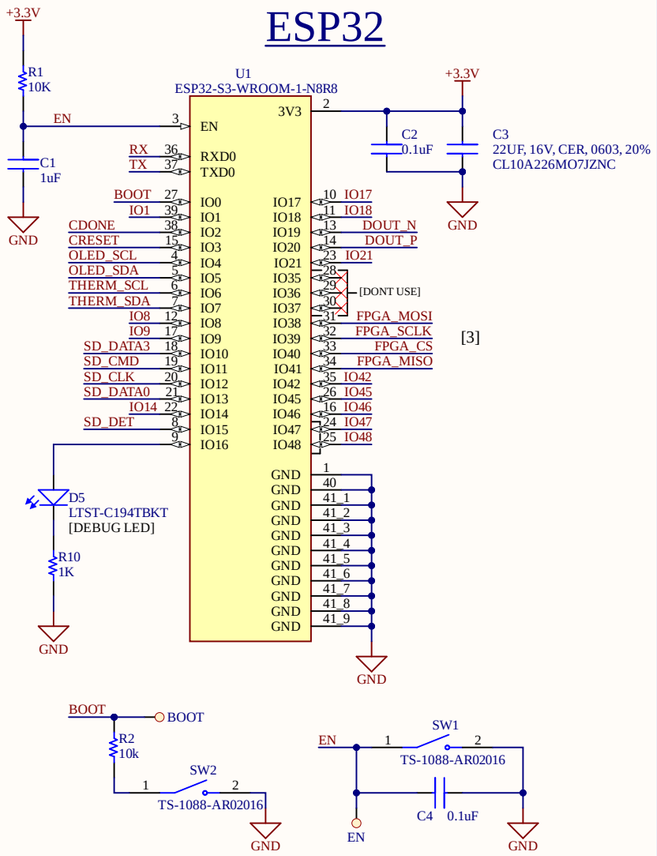
TITLE
THERMAL EDGE DETECTION

SIZE		DWG NO	REV
B		EE486S2026	-
SCALE: NONE	VIEWDRAW	SHEET 1 OF 3	

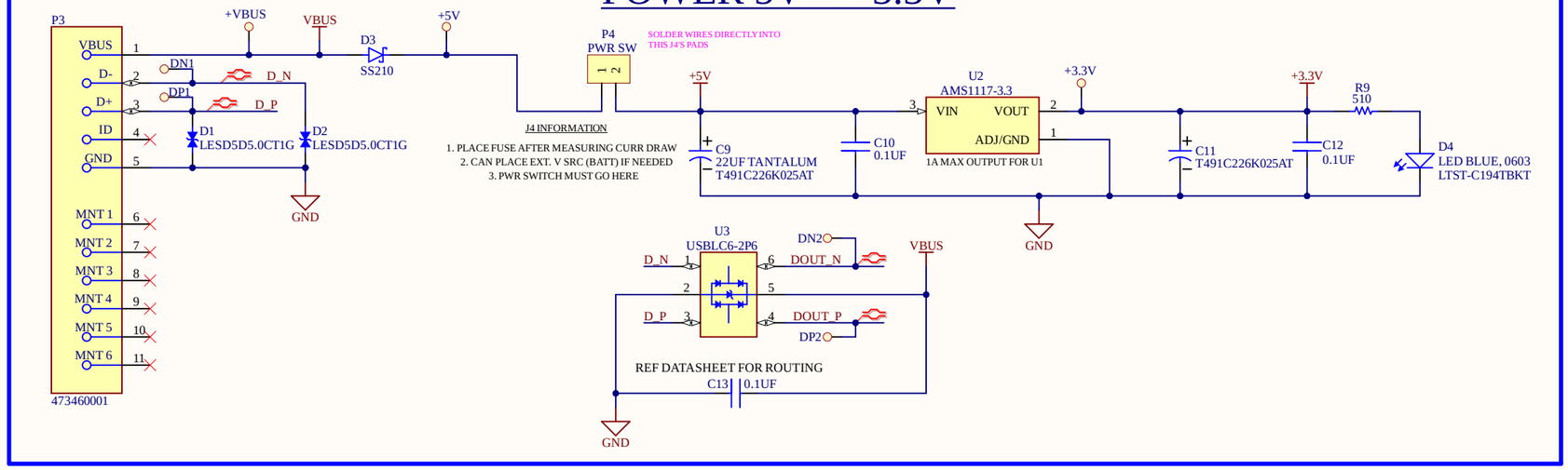
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

ESP32

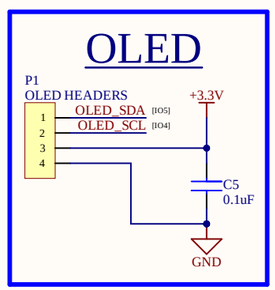
ESP32



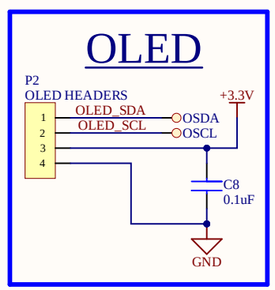
POWER 5V --> 3.3V



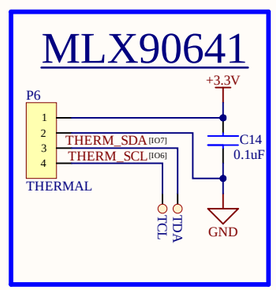
OLED



OLED



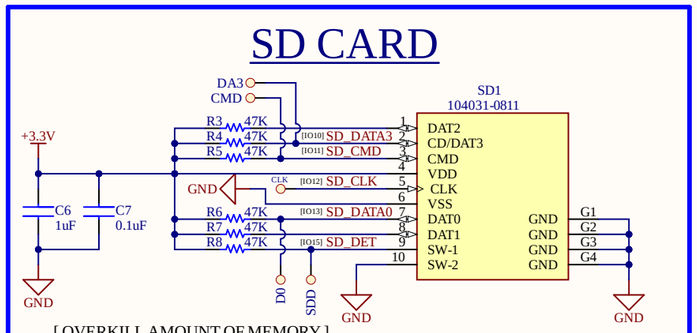
MLX90641



ADD SECOND OLED HEADER TO INCLUDE THE OPTION TO
HAVE 2 DISPLAYS/DOUBLE NIGHT VISION GOGGLES

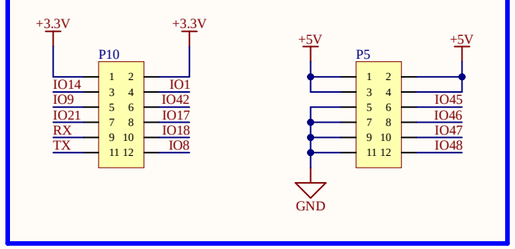
[SOLDER WIRES TO PADS]

SD CARD



[OVERKILL AMOUNT OF MEMORY]

ESP32 UNUSED PINS

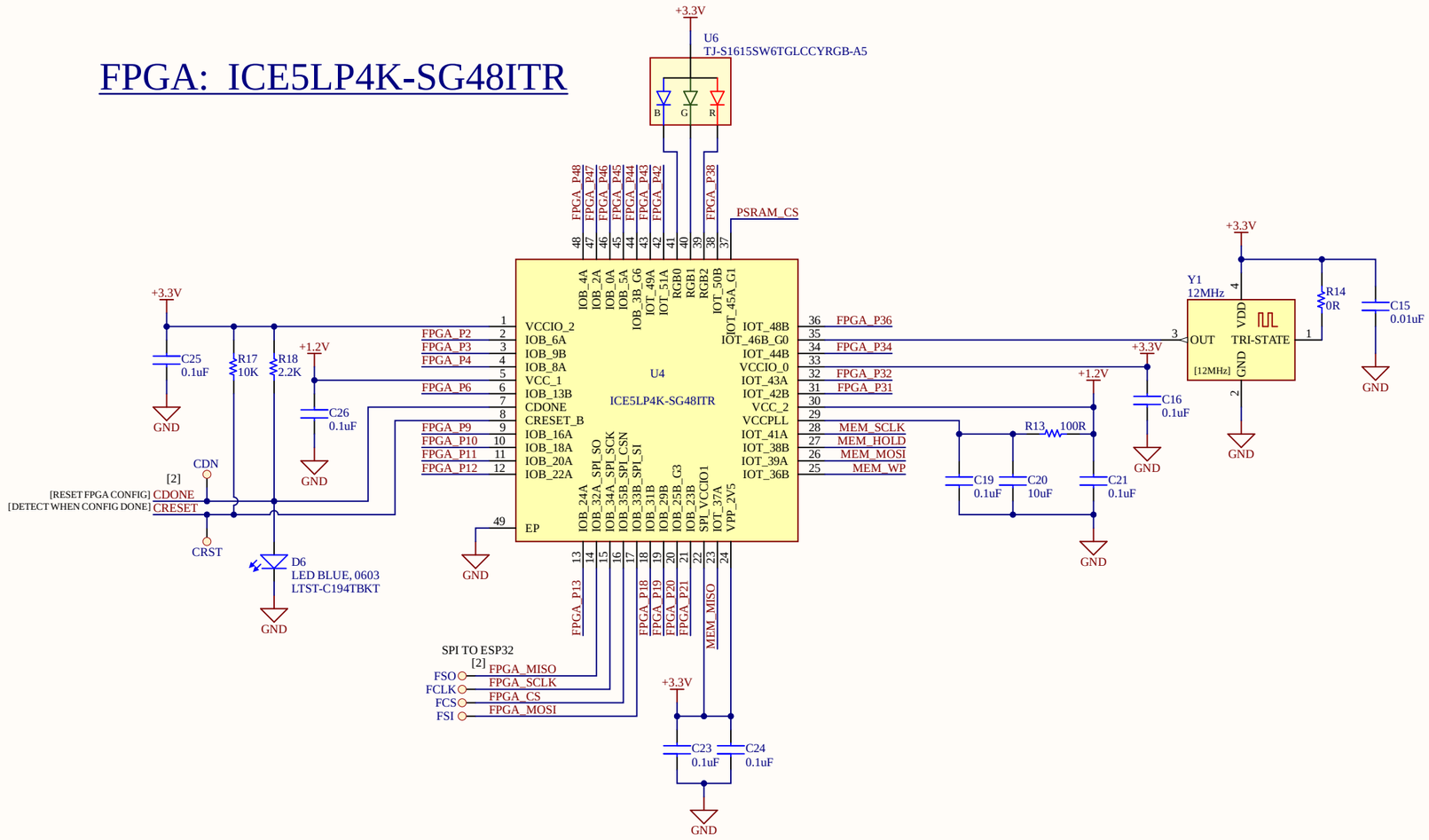


SIZE		DWG NO	REV
B		EE486S2026	-
SCALE: NONE	VIEWDRAW	SHEET 2 OF 3	

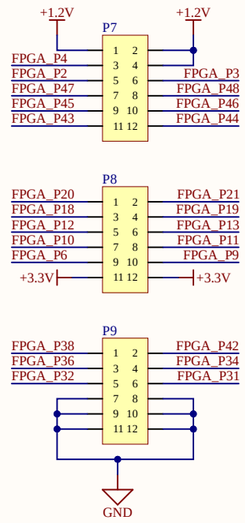
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

FPGA

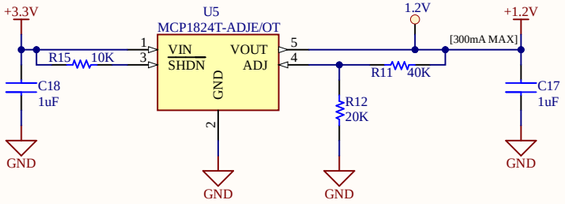
FPGA: ICE5LP4K-SG48ITR



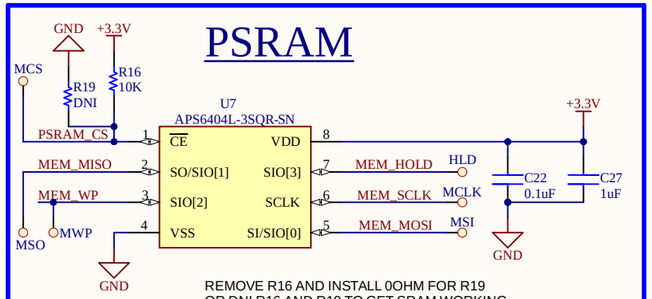
FPGA UNUSED PINS



3.3V --> 1.2V

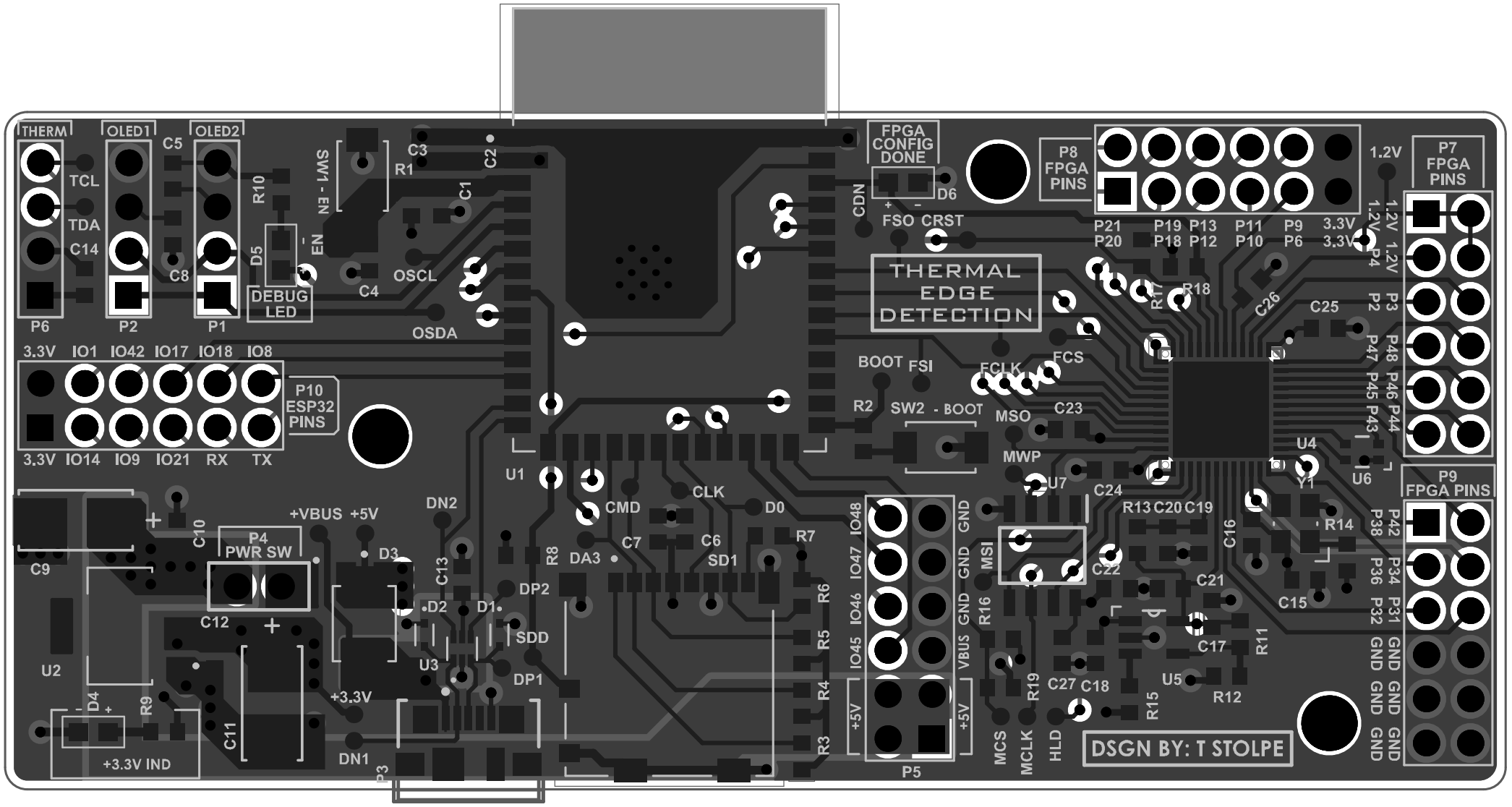


PSRAM



REMOVE R16 AND INSTALL 0OHM FOR R19
OR DNI R16 AND R19 TO GET SRAM WORKING

SIZE		DWG NO	REV
B		EE486S2026	-
SCALE: NONE	VIEWDRAW	SHEET 3 OF 3	


THERM
OLED1
OLED2
TCL
TDA
C14
C5
C8
R10
D5
DEBUG LED
SW1 - EN
EN
C3
R1
C2
C1
C4
OSCL
OSDA
P6
P2
P1
3.3V IO1 IO42 IO17 IO18 IO8
3.3V IO14 IO9 IO21 RX TX
P10 ESP32 PINS
FPGA CONFIG DONE
D6
FSO CRST
CDN
THERMAL EDGE DETECTION
P8 FPGA PINS
P21 P19 P13 P11 P9 3.3V
P20 P18 P12 P10 P6 3.3V
1.2V
P7 FPGA PINS
P4 P3 P2 P48 P47 P46 P45 P44 P43
R17
R18
C26
C25
BOOT
FSI
FCLK
FCS
R2
SW2 - BOOT
MSO
C23
MWP
U4
U7
U6
Y1
C24
R13 C20 C19
C16
R14
P9 FPGA PINS
P42 P38 P34 P36 P31 P32 GND GND GND GND GND GND
C22
C21
C15
C17
R11
U5
R12
C27 C18
R15
R19
MCS
MCLK
HLD
DSGN BY: T STOLPE
U1
CMD
CLK
D0
C7
C6
SD1
R7
R8
DA3
R6
R5
R4
R3
DP2
DP1
SDD
D2
D1
DN2
DN1
P3
U3
+VBUS +5V
P4 PWR SW
D3
C13
C12
C10
C9
U2
D4
R9
C11
+3.3V IND
+3.3V
IO48 IO47 IO46 IO45 GND GND GND VBUS +5V +5V
MSI
R16
P5